

EDITORIAL

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A cardiometabolic perspective on post-exertional malaise in myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) and Long COVID

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Abstract

Post-exertional malaise (PEM), the defining feature of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), is increasingly recognized in individuals with Long COVID. Although traditionally viewed as symptom exacerbation and/or emergence of new symptoms following exertion, PEM may reflect disrupted coordination across interconnected physiological systems operating over distinct post-exertional timescales. Such disruption may result in altered post-exertional physiological trajectories that contribute to multisystem manifestations and potential cardiometabolic consequences. This perspective outlines a conceptual approach for investigating PEM through longitudinal assessment of post-exertional physiological trajectories while considering disease severity and underlying cardiometabolic health, including obesity and type 2 diabetes. Integrating these dimensions may help contextualize post-exertional responses, inform future longitudinal cardiometabolic profiling, facilitate patient stratification, and provide a framework for future mechanistic and interventional studies in ME/CFS and Long COVID.

Keywords Cardiometabolic health, Post-exertional malaise, Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), Long COVID, Obesity, Type 2 diabetes

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From post-exertional homeostasis to post-exertional malaise

The physiological response to physical exertion depends on coordinated interactions across multiple organs and tissues. In healthy individuals, physical activity is accompanied by integrated cardiovascular, endothelial, metabolic, endocrine, autonomic, muscular and immune responses, together with dynamic regulation of energy production, substrate utilization, and fluid balance, all of which help maintain systemic homeostasis. These responses are normally coordinated across physiological systems and timescales, contributing to adaptive post-exertional trajectories that ultimately restore physiological homeostasis. Maintaining the integrity of these adaptive responses is fundamental to cardiometabolic homeostasis [1–4].

Within the spectrum of post-acute infection syndromes (PAIS) [5], post-exertional malaise (PEM) is the defining feature of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) [6] and is increasingly recognized in individuals with Long COVID [7, 8]. These two conditions share symptom profiles [9], providing an opportunity to investigate post-exertional pathophysiology [10]. Traditionally, PEM is defined as the worsening of symptoms following physical, cognitive, or emotional exertion [6]. Although PEM may be triggered by different forms of exertion, this perspective focuses primarily on physiological responses to physical exertion and the post-exertional trajectories that may contribute to the cardiometabolic manifestations of ME/CFS and Long COVID. The substantial overlap between the physiological systems involved in post-exertional adaptation and those implicated across the spectrum of cardiometabolic disease suggests that investigating post-exertional physiological trajectories may provide broader insights into cardiometabolic dysfunction across conditions such as obesity, metabolic syndrome, and type 2 diabetes (T2D).

From this perspective, PEM may reflect a broader disturbance in the coordination of multiple physiological systems rather than simply the consequence of exertion. While the mechanisms underlying PEM remain incompletely understood [10], one possibility is that physiological processes that normally operate in a coordinated manner become transiently or persistently dysregulated, contributing to altered post-exertional physiological trajectories across interconnected systems (Fig. 1). This interpretation is consistent with evidence of abnormalities involving cardiovascular, endothelial, metabolic, endocrine, autonomic, muscular and immune systems both in ME/CFS [11] and Long COVID [7], although the causality between these abnormalities and PEM remains uncertain. Viewed from this perspective, PEM may also provide a unique research opportunity to understand how cardiometabolic adaptation is achieved under

conditions of physiological stress and how underlying cardiometabolic health shapes post-exertional physiological trajectories.

PEM beyond symptom exacerbation

Notwithstanding its defining feature of both ME/CFS and Long COVID, PEM remains incompletely assessed and characterized in Long COVID cohorts [12]. Relatively few studies have incorporated pre- and post-exertional approaches to assess PEM directly [10]. As a result, PEM is typically viewed from a diagnostic or clinical perspective rather than as a pathophysiological process requiring mechanistic investigation.

Recent recommendations for Long COVID clinical trials further highlight the complexity of PEM as a pathophysiological phenomenon [13]. PEM may exhibit delayed onset following exertion and is characterized by substantial temporal variability, prolonged recovery, and, in some cases, incomplete return to pre-exertional baseline functional capacity. Consequently, the physiological state of a participant at the time of assessment may substantially influence the interpretation of study outcomes. This concept is supported by emerging cardiovascular and cardiometabolic evidence showing that physiological abnormalities in ME/CFS may become more apparent under orthostatic challenge than under resting conditions, indicating that the physiological context in which patients are evaluated can substantially influence the observed phenotype [14]. These recommendations also emphasize documenting whether participants are experiencing PEM during data collection, while recognizing that study procedures themselves may influence symptom expression during the assessment period [13, 15].

Importantly, the physiological impact of exertion is unlikely to be the same across all levels of disease severity. While individuals with mild or moderate disease may tolerate certain exertional assessment protocols, including cardiopulmonary exercise testing (CPET), hand-grip, or tilt-table testing [16–18], those who are severely affected or bed-bound often live with limited physiological reserve and may experience symptom worsening in response to challenges far less demanding than those imposed by conventional exertional testing [19]. In these patients, PEM may not simply reflect a more pronounced version of the response observed in milder cases, but a distinct physiological state characterized by altered post-exertional trajectories and reduced capacity to restore physiological homeostasis following exertion [20, 21]. These observations highlight the need to consider disease severity both when interpreting study findings and when selecting the timing and nature of physiological assessments.

This distinction becomes particularly relevant as the physiological systems involved in post-exertional

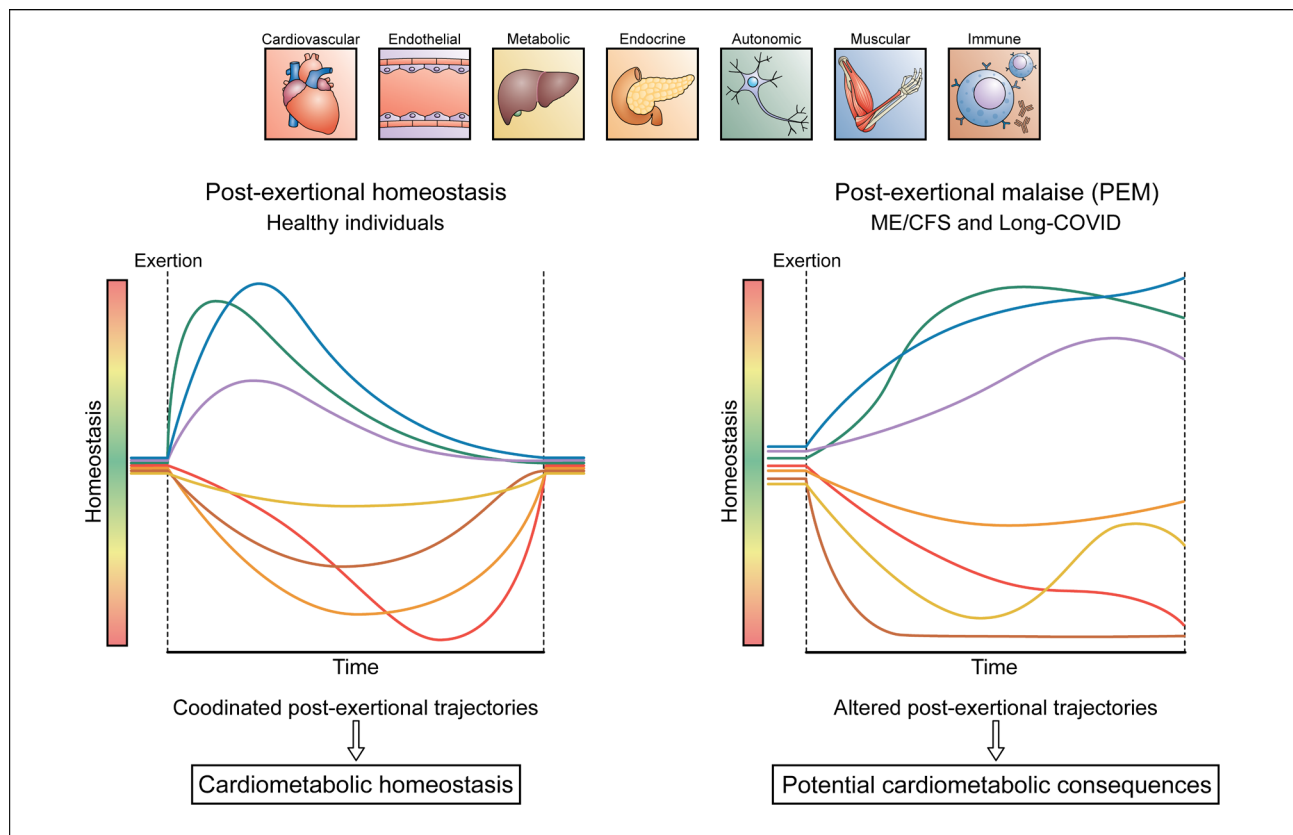


Fig. 1 Proposed cardiometabolic conceptual framework for coordinated and altered post-exertional physiological trajectories in health and post-exertional malaise (PEM) in myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) and Long COVID. The colored curves conceptually represent post-exertional physiological trajectories across cardiovascular, endothelial, metabolic, endocrine, autonomic, muscular, and immune systems rather than specific biological variables. The color scale reflects the homeostatic status of each physiological system, ranging from preserved (green) through altered (yellow) to impaired (red). In healthy individuals, coordinated post-exertional physiological trajectories facilitate the restoration of post-exertional and cardiometabolic homeostasis. In contrast, PEM is proposed to be characterized by altered post-exertional physiological trajectories associated with impaired coordination across physiological systems. The lower baseline and early divergence of trajectories in the PEM panel conceptually illustrate pre-existing alterations in multisystem coordination that become more apparent following exertion. Altered post-exertional physiological trajectories may contribute to impaired cardiometabolic homeostasis, cumulative cardiometabolic burden, and increased susceptibility to long-term cardiometabolic consequences

adaptation operate across different timescales. Some responses, including autonomic, hemodynamic, vascular, and endothelial adjustments, occur within seconds to minutes after exertion. Others unfold over hours to days and involve inflammatory signaling, hepatic and systemic metabolic regulation, substrate utilization, immune-metabolic interactions, and energetic processes. Longer-term alterations may further influence broader cardiometabolic regulation, potentially contributing to distinct cardiometabolic profiles following repeated exertional episodes [22, 23]. One possibility is that PEM reflects a disruption in the normal temporal coordination of physiological responses to physical exertion (Fig. 1). Such desynchronization may contribute both to the heterogeneity of responses observed across individuals with ME/CFS and Long COVID and to inconsistent findings when physiological responses are assessed at different stages of the post-exertional period.

This temporal complexity highlights the importance of longitudinal assessment. As post-exertional responses evolve at different rates across physiological domains [1–4], understanding PEM depends not only on what is measured, but also on when it is measured. Viewed from this perspective, the timing of assessment may become as important as the nature of the assessment itself [24, 25]. Together, these observations suggest that characterization of PEM depends not only on when physiological responses are measured, but also on the physiological context in which they are assessed. Integrating these temporal and physiological dimensions may therefore be critical for understanding the heterogeneity of PEM and identifying distinct post-exertional physiological trajectories across biologically and clinically meaningful patient subgroups.

A cardiometabolic interpretation of recurrent post-exertional dysregulation

Over the course of these chronic conditions, individuals with ME/CFS and Long COVID experience recurrent cycles of physical exertion, PEM, and prolonged physiological disturbance [6, 7, 10, 15]. The cumulative pathophysiological consequences of these repeated episodes remain largely unexplored. One possibility is that recurrent episodes of PEM are accompanied by disruption of the coordinated physiological responses required for post-exertional adaptation. Although individual episodes are often transient, repeated disruption of physiological system coordination could contribute to alterations in post-exertional physiological trajectories and cumulative disturbances in cardiometabolic regulation over time, potentially leading to distinct cardiometabolic profiles across individuals (Fig. 1).

A growing body of evidence implicates abnormalities involving endothelial function, oxygen extraction, autonomic regulation, lactate metabolism, coagulation pathways, metabolic flexibility, glucose homeostasis, and lipid metabolism both in ME/CFS [11] and Long COVID [7]. Reduced stroke volume, cardiac output, and cerebral blood flow, particularly during orthostatic challenge, have been consistently reported in ME/CFS, providing objective evidence of impaired cardiovascular adaptation [26]. Moreover, impaired capillary blood flow associated with endothelial dysfunction and vascular wall alterations has also been proposed as a contributor to impaired cardiovascular adaptation in ME/CFS and Long COVID [27]. Although these findings are often investigated separately, their frequent co-occurrence raises the possibility that they represent interconnected manifestations of a broader systems-level disturbance, rather than isolated abnormalities affecting individual physiological pathways.

If PEM reflects a disruption in the coordinated recovery of physiological systems following exertion, repeated episodes of post-exertional dysregulation could have consequences that extend beyond transient symptom worsening. Over time, recurrent disturbances across cardiovascular, autonomic, endothelial, metabolic, immune, and energetic pathways could contribute to a cumulative cardiometabolic burden, the magnitude of which may depend on the frequency, duration, and persistence of PEM episodes. Longitudinal cardiometabolic profiling may therefore help characterize post-exertional physiological trajectories and determine whether recurrent disturbances are associated with cumulative cardiometabolic burden, particularly across individuals with different cardiometabolic health profiles. Whether recurrent PEM contributes directly to such burden or instead reflects underlying pathophysiological processes that also influence disease severity, remains uncertain.

Importantly, immune, inflammatory, neurovascular, and viral persistence mechanisms proposed in ME/CFS and Long COVID are not mutually exclusive with the perspective proposed here. Rather, PEM may represent a physiological state in which diverse biological disturbances converge and become expressed through altered coordination across interconnected systems following exertion. If so, a major challenge may not be to identify a single mechanism underlying PEM, but to understand how multiple mechanisms interact across systems and timescales to shape post-exertional physiological trajectories, and whether these interactions define distinct cardiometabolic profiles with different clinical and biological manifestations.

Future clinical and therapeutic considerations

The potential influence of cardiometabolic health, including pre-existing or acquired obesity, metabolic syndrome, and T2D, remains underexplored in ME/CFS and Long COVID research. These observations raise the possibility that underlying cardiometabolic health contributes to distinct post-exertional physiological trajectories rather than simply representing a clinical comorbidity. Greater inclusion of individuals across different cardiometabolic health profiles may provide important insights into how pre-existing cardiometabolic conditions influence post-exertional trajectories and contribute to the heterogeneity of PEM. Indeed, the relationship between Long COVID and T2D has emerged as an area of increasing research interest, with ongoing nationwide studies specifically designed to evaluate its epidemiology and long-term health consequences [28]. However, cardiometabolic health is unlikely to be the only source of variability influencing post-exertional responses.

Beyond cardiometabolic health, factors such as age, biological sex, disease duration, and disease severity are also likely to influence both post-exertional physiological responses and the trajectories of post-exertional symptoms. Rather than simply modifying the magnitude of PEM, these characteristics may shape distinct post-exertional physiological trajectories across systems and timescales. Future studies should therefore consider these dimensions in both study design and data interpretation to improve patient stratification and identify biologically and clinically meaningful patient subgroups.

A better understanding of these sources of variation may also help inform the evaluation of cardiometabolic interventions in PAIS [29]. Moreover, the growing interest in repurposing established cardiometabolic therapies, including metformin, reflects an evolving view that cardiometabolic pathways may represent potential therapeutic targets in ME/CFS and Long COVID [30]. Among emerging cardiometabolic therapies, glucagon-like peptide-1 receptor agonists (GLP-1RAs) are also of

particular interest because of their pleiotropic effects across cardiovascular, metabolic, endothelial, autonomic, inflammatory, and neuroimmune pathways [31], several of which have been implicated in both ME/CFS [11] and Long COVID [7]. Ongoing initiatives, including the NIH RECOVER-TLC program [32] and the LoCITT-T trial [33], are beginning to explore this therapeutic space. Whether such interventions influence PEM remains unknown. However, interventions capable of modulating multiple physiological systems simultaneously and improving cardiometabolic function may not only offer therapeutic potential but also provide an opportunity to determine whether improving cardiometabolic function modifies post-exertional physiological trajectories and ultimately influences the manifestation and heterogeneity of PEM.

Concluding remarks

PEM has traditionally been viewed as symptom exacerbation and/or emergence of new symptoms following exertion. This perspective proposes that PEM may instead reflect disrupted coordination across interconnected physiological systems operating over distinct post-exertional timescales. Viewed from this perspective, recurrent post-exertional disturbances may contribute not only to symptom burden, but also to altered cardiometabolic adaptation and broader cardiometabolic consequences in ME/CFS and Long COVID. Rather than viewing PEM solely through a clinical lens, this perspective proposes that PEM provides an opportunity to investigate how physiological systems coordinate and adapt over time following physiological stress. Integrating longitudinal physiological assessment with longitudinal cardiometabolic profiling may therefore help identify distinct post-exertional physiological trajectories, improve patient stratification, and generate new hypotheses regarding the mechanisms linking post-exertional dysregulation to cardiometabolic manifestations in ME/CFS and Long COVID.

Key questions

- Does PEM reflect disrupted coordination across physiological systems and timescales?
- Can distinct post-exertional physiological trajectories define patient subgroups?
- Does recurrent post-exertional physiological dysregulation contribute to cumulative cardiometabolic burden?
- How do the timing of assessment and disease heterogeneity influence the identification of post-exertional physiological trajectories?

- Can longitudinal cardiometabolic profiling identify biomarkers associated with distinct post-exertional trajectories?

Together, addressing these questions may help determine whether PEM represents disruption of the coordinated physiological responses that shape post-exertional physiological trajectories, and how these alterations contribute to the multisystem and cardiometabolic manifestations of ME/CFS and Long COVID. Ultimately, advancing our understanding of PEM may require moving beyond symptom characterization toward the longitudinal investigation of post-exertional physiological trajectories across interconnected physiological systems and timescales, while integrating cardiometabolic health as an important source of biological heterogeneity, particularly in individuals with obesity and T2D, in future mechanistic and interventional studies.

Abbreviations

CPET	cardiopulmonary exercise testing
GLP-1RA	glucagon-like peptide-1 receptor agonist
ME/CFS	myalgic encephalomyelitis/chronic fatigue syndrome
PAIS	post-acute infection syndromes
PEM	post-exertional malaise
T2D	type 2 diabetes

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Author contributions

FW conceived the editorial, developed the conceptual framework, and wrote the first draft of the manuscript. EP, EU, RB, NS, and EF contributed to the conceptual development of the perspective, critically revised the manuscript, and approved the final version.

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Declarations

Competing interests

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